Omkar Rana

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Summary

Data Engineer with 3+ years of experience building cloud data pipelines using Python, PySpark, AWS Glue, and Lambda. Hands-on experience in Machine Learning, NLP, and Generative AI, with a focus on building data systems that support ML workflows.

Skills

Languages: Python, SQL, Java, C/C++

Cloud & Big Data: AWS (Glue, Lambda, SageMaker, EC2, CDK), Databricks, Azure, PySpark, Airflow, Delta Lake

AI/ML & GenAI: LLMs, RAG, LangChain, TensorFlow/PyTorch

MLOps/LLMOps: MLflow, CI/CD, Docker, Kubernetes, Model Monitoring and Observability

Databases & Tools: Git, Vector DBs (Pinecone/FAISS)

Experience

Tata Consultancy Services — Gandhinagar, Gujarat

Data Engineer — *Nov 2023 – Present*

- Built PySpark pipelines on AWS Glue and Lambda to prepare healthcare data for ML-driven recommendations
- Worked on database migration to reduce the data latency for downstreams
- Trained 50+ associates in Python and AWS, aligning their skills with project requirements

ESTEE Advisors — Gandhinagar, Gujarat

Data Analyst — *Sep 2023 – Nov 2023*

- Built visual reports for trading results to help the team analyze market patterns
- Improved Python scripts used to run trading strategies automatically on schedule

Projects

Label Detection | *Deep Learning, OpenCV, Python*

- Extract the label number from the side of cement bags from a video which has bags moving over conveyor belt
- Used OpenCV to convert the video to images and detect the bags
- Calculated average time each bag spent on screen to optimise the process and improve accuracy to over 93%

Parking Aid System | *OpenCV, MATLAB*

- Displays an image of parking lot at the entrance to identify empty parking slots
- Using OpenCV and other MATLAB libraries, detected open spaces from the top view image of parking lot

Certifications

Databricks Certified Data Engineer Associate ↗ 167910451

Databricks Certified Generative AI Engineer Associate ↗ 176997960

AWS AI Practitioner ↗ f643abaea337438bbb6d864e144815df

Claude Certified Architect - Foundations ↗

Education

Nirma University — Ahmedabad, Gujarat

Bachelor of Technology in Computer Science and Engineering — *Jul 2019 – May 2023*